

Exercise 1

Schema

```
Emp(eid:integer, ename:string, age:integer, salary:real)
Works(eid:integer, did:integer, pct_time:integer)
Dept(did:integer, dname:string, budget:real, managerid:integer)
```

Note

- Works.*pct_time* is the employee percent working time. It is expressed as a number between 0 (not working) and 100 (full-time).

Queries

1. Print the names and ages of each employee who works in both the Hardware department and the Software department.
2. For each department with more than 20 full-time-equivalent employees (i.e., where the part-time and full-time employees add up to at least that many full-time employees), print the *did* together with the number of employees that work in that department.
3. Print the name of each employee whose salary exceeds the budget of all of the departments that he or she works in.
4. Find the *managerids* of managers who manage only departments with budgets greater than \$1 million.
5. Find the *enames* of managers who manage the departments with the largest budgets.
6. If a manager manages more than one department, he or she *controls* the sum of all the budgets for those departments. Find the *managerids* of managers who control more than \$5 million.
7. Find the *managerids* of managers who control the largest amounts.
8. Find the *enames* of managers who manage only departments with budgets larger than \$1 million, but at least one department with budget less than \$5 million.

Solutions exercise 1

1.

```
SELECT E.ename, E.age
FROM Emp E, Works W1, Works W2, Dept D1, Dept D2
WHERE E.eid=W1.eid AND W1.did=D1.did AND D1.dname='Hardware'
      AND E.eid=W2.eid AND W2.did=D2.did AND D2.dname='Software'
```
2.

```
SELECT did, COUNT(eid)
FROM Works
GROUP BY did
HAVING 2000 < SUM(pct_time);
```
3.

```
SELECT ename
FROM Emp E
WHERE salary > ALL(
    SELECT budget
    FROM Dept D, Works W
    WHERE E.eid = W.eid AND D.did = W.did
);
```
4.

```
SELECT managerid
FROM Dept
GROUP BY mangerid
HAVING MIN(budget) > 1000000;
```

— Alternative

```
SELECT DISTINCT D.managerid
FROM Dept D
WHERE 1000000 < ALL(
    SELECT D2.budget
    FROM Dept D2
    WHERE D2.managerid=D.managerid
);
```
5.

```
SELECT ename
FROM Emp
WHERE eid IN (
    SELECT managerid
    FROM Dept
    WHERE budget=(
        SELECT MAX(budget)
        FROM Dept
    )
);
```
6.

```
SELECT managerid
FROM Dept
GROUP BY mangerid
HAVING SUM(budget) > 5000000;
```

— Alternative

```
SELECT D.managerid FROM Dept D
WHERE 5000000 < (
    SELECT SUM(D2.budget)
    FROM Dept D2
    WHERE D2.managerid=D.managerid
);
```

7. `SELECT DISTINCT` managerid
`FROM` (
 `SELECT DISTINCT` managerid , `SUM`(budget) `AS` tot_budget
 `FROM` Dept
 `GROUP BY` managerid
)
`AS` control
`WHERE` control.tot_budget=(
 `SELECT MAX`(tot_budget)
 `FROM` control
);
8. `SELECT` ename
`FROM` Emp E
`JOIN` Dept D `ON` eid = managerid
`GROUP BY` eid , ename
`HAVING` `EVERY`(budget > 1000000)
 `AND ANY`(budget < 5000000)

Exercise 2

Schema

```
Flights(flno:integer, from:string, to:string, distance:integer,  
        departs:time, arrives:time, price:real)  
Aircraft(aid:integer, aname:string, cruising_range:integer)  
Certified(eid:integer, aid:integer)  
Employees(eid:integer, ename:string, salary:integer)
```

Notes

- A pilot is an employee certified on at least one aircraft.
- An example of Aircraft *aname* is 'Boing 747'.

Queries

1. Find the names of aircraft such that all pilots certified to operate them have salaries more than \$80,000.
2. For each pilot who is certified for more than three aircraft, find the *eid* and the maximum cruising range of the aircraft for which she or he is certified.
3. Find the names of pilots whose *salary* is less than the price of the cheapest route from Los Angeles to Honolulu.
4. For all aircraft with cruising range over 1000 miles, find the name of the aircraft and the average salary of all pilots certified for this aircraft.
5. Find the names of pilots certified for some Boeing aircraft.
6. Find the *aids* of all aircraft that can be used on routes from Los Angeles to Chicago.
7. Identify the routes that can be piloted by every pilot who makes more than \$100,000 .
8. Print the *enames* of pilots who can operate planes with cruising range greater than 3000 miles but are not certified on any Boeing aircraft.
9. A customer wants to travel from Madison to New York with no more than two changes of flight. List the choice of departure times from Madison if the customer wants to arrive in New York by 6 p.m.
10. Compute the difference between the average salary of a pilot and the average salary of all employees (including pilots).
11. Print the name and salary of every non-pilot whose salary is more than the average salary for pilots.
12. Print the names of employees who are certified only on aircrafts with cruising range longer than 1000 miles.
13. Print the names of employees who are certified only on aircrafts with cruising range longer than 1000 miles, but on at least two such aircrafts.
14. Print the names of employees who are certified only on aircrafts with cruising range longer than 1000 miles and who are certified on some Boeing aircraft.

Solutions exercise 2

1.

```
SELECT aname
FROM Aircraft
WHERE aid NOT IN (
    SELECT aid
    FROM Certified
    NATURAL JOIN Employee
    WHERE salary < 80000
);
```
2.

```
SELECT eid, MAX(A.cruising_range)
FROM Certified
NATURAL JOIN Aircraft
GROUP BY eid
HAVING COUNT(*) > 3
```
3.

```
SELECT DISTINCT ename
FROM Employees
NATURAL JOIN Certified
WHERE salary < (
    SELECT MIN(price)
    FROM Flights
    WHERE "from"='Los_Angeles' AND to='Honolulu'
);
```
4. — NOTE: We are grouping also by 'aname' because each select list element must either appear in the GROUP BY statement, or be the result of an aggregation function, or be a constant.

```
SELECT aname, AVERAGE(salary)
FROM Aircraft
NATURAL JOIN Certified
NATURAL JOIN Employees
WHERE cruising_range > 1000
GROUP BY aid, aname
```
5.

```
SELECT DISTINCT ename
FROM Employees
NATURAL JOIN Certified
NATURAL JOIN Aircraft
WHERE aname LIKE 'Boeing%';
```
6.

```
SELECT aid
FROM Aircraft
WHERE cruising_range > (
    SELECT MIN(distance)
    FROM Flights
    WHERE "from"='Los_Angeles' AND to='Chicago'
);
```
7.

```
SELECT "from", to
FROM Flights
WHERE distance <= (
    SELECT MIN(max_range)
    FROM (
        SELECT MAX(cruising_range) AS max_range
        FROM Aircraft
        NATURAL JOIN Certified
        NATURAL JOIN Employee
        WHERE salary > 100000
        GROUP BY eid
    )
);
```

```

))

-- Alternative
SELECT DISTINCT "from", to
FROM Flights
WHERE NOT EXISTS (
    SELECT *
    FROM Employees E
    WHERE salary > 100000 AND NOT EXISTS (
        SELECT *
        FROM Aircraft A, Certified C
        WHERE cruising_range > distance
            AND E.eid = C.eid
            AND A.aid = C.aid))

8. SELECT ename
   FROM Employees
  NATURAL JOIN Certified
  NATURAL JOIN Aircraft
  GROUP BY eid, ename
  HAVING ANY(cruising_range > 3000)
        AND NOT ANY(aname LIKE 'Boeing%');

9. SELECT departs
   FROM Flights
  WHERE flno IN (
    -- direct
    (SELECT flno
     FROM Flights
    WHERE "from"='Madison' AND to='New_York'
      AND FO.arrives < '18:00')
    UNION
    -- 2 legs
    (SELECT FO.flno
     FROM Flights FO, Flights F1
    WHERE FO."from"='Madison' AND FO.to<>'New_York'
      AND FO.to=F1."from" AND F1.to='New_York'
      AND F1.departs>FO.arrives AND F1.arrives<'18:00')
    UNION
    -- 3 legs
    (SELECT FO.flno
     FROM Flights FO, Flights F1, Flights F2
    WHERE FO.to=F1."from" AND F1.to=F2."from"
      AND FO."from"='Madison' AND F2.to='New_York'
      AND FO.to<>'New_York' AND F1.to<>'New_York'
      AND F1.departs>FO.arrives AND F2.departs>F1.arrives
      AND F2.arrives < 18:00')
  );

10. SELECT pilots.avg - "all".avg
   FROM (
     SELECT AVG(salary) AS avg
     FROM Employees
     WHERE eid IN (
       SELECT DISTINCT eid
       FROM Certified C
     )
   ) AS pilots, (
     SELECT AVG(salary) AS avg
     FROM Employees
   ) AS "all";

```

11.

```
SELECT ename, salary
FROM Employees
WHERE eid NOT IN (
    SELECT DISTINCT eid
    FROM Certified
) AND E.salary > (
    SELECT AVG (salary)
    FROM Employees
    WHERE eid IN (
        SELECT DISTINCT eid
        FROM Certified
    )
);
```
12.

```
SELECT ename
FROM Employees
NATURAL JOIN Certified
NATURAL JOIN Aircraft
GROUP BY eid, ename
HAVING EVERY(cruising_range > 1000);
```
13.

```
SELECT ename
FROM Employees
NATURAL JOIN Certified
NATURAL JOIN Aircraft
GROUP BY eid, ename
HAVING EVERY(cruising_range > 1000)
    AND COUNT(aid) >= 2;
```
14.

```
SELECT ename
FROM Employees
NATURAL JOIN Certified
NATURAL JOIN Aircraft
GROUP BY eid, ename
HAVING EVERY(cruising_range > 1000)
    AND ANY (aname LIKE 'Boeing%');
```